Summary

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SUMMARY: Velocity, Speed, Position, Displacement, Distance

Position, $s(t)$, of an object at time t:

$$s(t) = s(a) + \int_a^t v(z)dz$$

Displacement, $\Delta s = s(b) - s(a)$,
of an object over the time interval $[a, b]$:

$$s(b) - s(a) = \int_a^b v(t)dt$$

Distance traveled by an object
over the time interval $[a, b]$:

$$\int_a^b |v(t)|dt$$

SUMMARY: Rate of Accumulation, Amount, Change in the Amount

The amount, $A(t)$, of some
substance/population at the time t:

$$A(t) = A(a) + \int_a^t A'(z)dz$$

The amount, $A(b)$,
over the time interval $[a, b]$:

$$A(b) = A(a) + \int_a^b A'(t)dt$$

The change in the amount, $A(b) - A(a)$,
over the time interval $[a, b]$:

$$A(b) - A(a) = \int_a^b A'(t)dt$$

SUMMARY: Average Value, the Mean Value Theorem for Integrals

Average value $\bar{f}$,

of the function f on the interval $[a, b]$: $$\bar{f} = \frac{1}{b-a}\int_a^b f(x)dx$$

Mean Value Theorem for Integrals

Let f be continuous on $[a, b]$. There exists a value c in (a, b) such that

$$\frac{1}{b-a}\int_a^b f(x)dx = f(c)$$

Note 1: $f(c) = \bar{f}$, the average value of f on $[a, b]$.

Note 2: The net area of the region between the curve $y = f(x)$ and the $x-$axis is given by

$$\int_a^b f(x)dx = f(c)(b-a)$$